

An engineering solution for mesh size effects in the simulation of delamination using cohesive zone models

A. Turon ^a, C.G. Dávila ^b, P.P. Camanho ^{c,*}, J. Costa ^a

^a *AMADE, Polytechnic School, University of Girona, Campus Montilivi s/n, 17071 Girona, Spain*

^b *NASA Langley Research Center, Hampton, VA, USA*

^c *DEMEGI, Faculdade de Engenharia, Universidade do Porto, Rua Dr. Roberto Frias, 4200-465 Porto, Portugal*

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Abstract

A methodology to determine the constitutive parameters for the simulation of progressive delamination is proposed. The procedure accounts for the size of a cohesive finite element and the length of the cohesive zone to ensure the correct dissipation of energy. In addition, a closed-form expression for estimating the minimum penalty stiffness necessary for the constitutive equation of a cohesive finite element is presented. It is shown that the resulting constitutive law allows the use of coarser finite element meshes than is usually admissible, which renders the analysis of large-scale progressive delamination problems computationally tractable.

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1. Introduction

Delamination, or interfacial cracking between composite layers, is one of the most common types of damage in laminated fibre-reinforced composites due to their relatively weak interlaminar strengths. Delamination may arise under various circumstances, such as low velocity impacts, or bearing loads in structural joints. The delamination failure mode is particularly important for the structural integrity of composite structures because it is difficult to detect during inspection.

The simulation of delamination using the finite element method (FEM) is normally performed by means of the Virtual Crack Closure Technique (VCCT) [1], or using cohesive finite elements [2–12].

The VCCT is based on the assumption that the energy released during delamination propagation equals the work required to close the crack back to its original position. Based on this assumption, the components of the energy release rate are computed from the nodal forces and nodal relative displacements [1]. Delamination

* Corresponding author. Tel.: +351 22 5081753; fax: +351 22 5081584.

E-mail address: pcamanho@fe.up.pt (P.P. Camanho).

growth is predicted when a combination of the components of the energy release rate is equal to a critical value.

There are some difficulties when using the VCCT in the simulation of progressive delamination. The calculation of fracture parameters requires nodal variables and topological information from the nodes ahead and behind the crack front. Such calculations are tedious to perform and may require remeshing for crack propagation.

The use of cohesive finite elements can overcome some of these above difficulties. Cohesive finite elements can predict both the onset and the non-self-similar propagation of delamination. However, the simulation of progressive delamination using cohesive elements poses numerical difficulties related to the proper definition of the stiffness of the cohesive layer, the requirement of extremely refined meshes, and the convergence difficulties associated with problems involving softening constitutive models.

A procedure to determine the optimal value of the penalty stiffness of a cohesive element is proposed. A closed-form expression for the interface stiffness is developed, replacing the empirical definitions of the stiffness of the cohesive layer that are normally used. In addition, a solution that allows the use of coarse meshes in the simulation of delamination is proposed. The procedure is based on an artificial increase of the cohesive zone length that it is obtained by lowering the interfacial strength while keeping the fracture toughness constant. It is shown that it is possible to predict the propagation of delamination accurately in specimens with and without pre-existing cracks by using coarse meshes and their correspondingly adjusted constitutive models.

The methodologies proposed here are relevant for the use of cohesive zone models in the simulation of delamination in large composite structures used in the industry where the requirements of extremely fine meshes cannot be met.

The kinematics and constitutive model of a previously proposed cohesive element [11,12], formulated for three-dimensional (3D) elements, is adapted to be used with two-dimensional (2D) finite element models. Several simulations of specimens with and without initial cracks are performed to demonstrate that the methodology proposed can accurately predict both crack initiation and propagation using coarse meshes.

2. Selection of cohesive zone model parameters

2.1. Cohesive zone model approach

Cohesive finite elements are used to model material discontinuities. Consider a domain Ω containing a material discontinuity, Γ_d which divides the domain Ω into two parts, Ω_+ and Ω_- . Prescribed tractions, t_i , are imposed on the boundary Γ_F . The stress field inside the domain, σ_{ij} , is related to the external loading and to the closing tractions τ_j^+ , τ_j^- in the material discontinuity through the equilibrium equations:

$$\sigma_{ij,j} = 0 \quad \text{in } \Omega \quad (1)$$

$$\sigma_{ij}n_j = t_i \quad \text{on } \Gamma_F \quad (2)$$

$$\sigma_{ij}n_j^+ = \tau_i^+ = -\tau_i^- = \sigma_{ij}n_j^- \quad \text{on } \Gamma_d \quad (3)$$

The formulation of the cohesive finite elements is based in the Cohesive Zone Model (CZM) approach. The CZM approach [13–15] is one of the most commonly used tools to investigate interfacial fracture. The CZM approach assumes that a cohesive damage zone develops near the crack tip. The CZM links the microstructural failure mechanism to the continuum fields governing bulk deformations. Thus, a CZM is characterized by the properties of the bulk material, the crack initiation condition, and the crack evolution function.

Cohesive damage zone models relate cohesive surface tractions, τ , to displacement jumps, Δ , at an interface where a crack may occur. Damage initiation is related to the interfacial strength, τ^0 , i.e., the maximum traction on the traction–displacement jump relation. The area under the traction–displacement jump relation is equal to the fracture toughness, G_c . There are several types of constitutive equations used in cohesive elements: Tvergaard and Hutchinson [16] proposed a trapezoidal law, Cui and Wisnom [17] a perfectly plastic rule, Needleman first proposed a polynomial law, [18], and later an exponential law, Xu and Needleman [19]. Goyal et al. [10] adopted Needleman's exponential law to account for load reversal without restoration of the

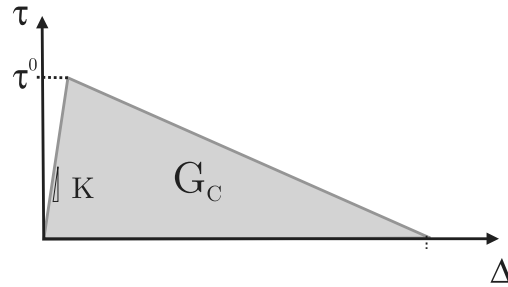


Fig. 1. Bilinear constitutive equation.

cohesive state. The law used in this paper is a bilinear relation between the tractions and the displacement jumps [6,9,11,12,20] (see Fig. 1). The slope of the constitutive equation before damage initiation, K , is referred as the interface stiffness.

2.2. Cohesive zone model and FEM

In a finite element model using the CZM approach, the complete material description is separated into fracture properties captured by the constitutive model of the cohesive surface and the properties of the bulk material, captured by the continuum regions.

To obtain a successful FEM simulation using CZM [21], two conditions must be met: (a) The cohesive contribution to the global compliance before crack propagation should be small enough to avoid the introduction of a fictitious compliance to the model [22,23], and (b) the element size must be less than the cohesive zone length [24,25].

2.2.1. Stiffness of the cohesive zone model

Different guidelines have been proposed for selecting the stiffness of the interface. Daudeville et al. [26] calculated the stiffness in terms of the thickness and the elastic modulus of the interface. The resin-rich interface between plies is of the order of 10^{-5} m. Therefore, the interface stiffness obtained from the Daudeville et al. equations is very high. Zou et al. [27], based on their own experience, proposed a value for the interface stiffness between 10^4 and 10^7 times the value of the interfacial strength per unit length. Camanho et al. [9] obtained accurate predictions for graphite/epoxy specimens using a value of 10^6 N/mm³.

The effective elastic properties of the whole laminate depend on the properties of both the cohesive surfaces and the bulk constitutive relations of the plies. Although the compliance of the cohesive surfaces can contribute to the global deformation, its only purpose is to simulate fracture. Moreover, the elastic properties of the cohesive surfaces are mesh-dependent because the surface relations exhibit an inherent length scale that is absent in homogeneous deformations [29].

If the cohesive contribution to the compliance is not small enough compared to that of the volumetric constitutive relation, a stiff connection between two neighboring layers before delamination initiation is not assured. The effect of compliance of the interface on the bulk properties of a laminate is illustrated in the 1D model shown in Fig. 2. The equilibrium condition requires

$$\sigma = E_3 \varepsilon = K \Delta \quad (4)$$

where σ is the traction on the surface, t is the thickness of an adjacent sub-laminate, $\varepsilon = \frac{\delta t}{t}$ is the transverse strain, K is the interface stiffness that relates the resulting tractions at the interface with the opening displacement Δ , and E_3 is the through-the-thickness Young's modulus of the material. For a transversely isotropic material $E_3 = E_2$.

The effective strain of the composite is

$$\varepsilon_{\text{eff}} = \frac{\delta t}{t} + \frac{\Delta}{t} = \varepsilon + \frac{\Delta}{t} \quad (5)$$

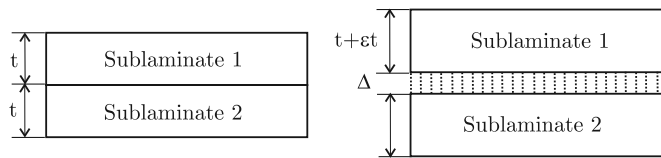


Fig. 2. Influence of the cohesive surface on the deformation.

Since the equilibrium condition requires that $\sigma = E_{\text{eff}} \epsilon_{\text{eff}}$, the equivalent Young's modulus E_{eff} can be written as a function of the Young's modulus of the material, the mesh size, and the interface stiffness. Using Eqs. (4) and (5), the effective Young's modulus can be written as

$$E_{\text{eff}} = E_3 \left(\frac{1}{1 + \frac{E_3}{Kt}} \right) \quad (6)$$

The effective elastic properties of the composite will not be affected by the cohesive surface whenever $E_3 \ll Kt$, i.e.,

$$K = \frac{\alpha E_3}{t} \quad (7)$$

where α is a parameter much larger than 1 ($\alpha \gg 1$). However, large values of the interface stiffness may cause numerical problems, such as spurious oscillations of the tractions [4]. Thus, the interface stiffness should be large enough to provide a reasonable stiffness but small enough to reduce the risk of numerical problems such as spurious oscillations of the tractions in an element. For values of α greater than 50, the loss of stiffness due to the presence of the interface is less than 2%, which is sufficiently accurate for most problems.

The use of Eq. (7) is preferable to the guidelines presented in previous works [9,26,27] because it results from mechanical considerations, and the resulting value of K does not significantly affect the compliance of the composite.

The spurious oscillations of the tractions resulting from an excessively stiff interface and Gauss integration scheme [28] are avoided by calculating the interface stiffness using Eq. (7), using linear shape functions and Newton–Cotes integration scheme [28,11,12].

The values of the interface stiffness obtained with Eq. (7) and those used by other authors for a carbon/epoxy composite are shown in Table 1. The material's transverse modulus is 11 GPa, its nominal interfacial strength is $\tau^0 = 45$ MPa, and α is taken as 50.

Eq. (7) gives a range of the interface stiffness between 10^5 and 5×10^6 N/mm³ for a sub-laminate thickness between 0.125 mm and 5 mm. These values are close to the interface stiffness proposed by Camanho and Dávila and the values obtained from Zou's guidelines (between 4.5×10^5 and 4.5×10^8 N/mm³).

2.2.2. Length of the cohesive zone

Under single-mode loading, interface damage initiates when the traction reaches the maximum nominal interfacial strength, τ^0 . For mixed-mode loading, damage onset is predicted by a criterion established in terms of the normal and shear tractions. Crack propagation occurs when the energy release rate reaches a critical value G_c . The CZM approach prescribes the interfacial normal and shear tractions that resist separation and relative sliding at an interface. The tractions, integrated to complete separation, yield the fracture energy

Table 1

Interface stiffness K proposed by different authors (N/mm³) and those calculated from Eq. (7)

t (mm)	0.125	1	2	3	5
Eq. (7)	4.43×10^6	5.5×10^5	2.75×10^5	1.83×10^5	1.1×10^5
Zou et al. [27]		4.5×10^5	$\leq K \leq$	4.5×10^8	
Camanho et al. [9]	10^6	10^6	10^6	10^6	10^6

release rate, G_c . The length of the cohesive zone l_{cz} is defined as the distance from the crack tip to the point where the maximum cohesive traction is attained.

Different models have been proposed in the literature to estimate the length of the cohesive zone. Irwin [30] estimated the size of the plastic zone ahead of a crack in a ductile solid by considering the crack tip zone where the von Mises equivalent stress exceeds the tensile yield stress. Dugdale [13] estimated the size of the yield zone ahead of a mode I crack in a thin plate of an elastic–perfectly plastic solid by idealizing the plastic region as a narrow strip extending ahead of the crack tip that is loaded by the yield traction. Barenblatt [14] provided an analogue for ideally brittle materials of the Dugdale plastic zone analysis. Hui et al. [31] estimated the length of the cohesive zone for soft elastic solids, and Falk et al. [21] and Rice [32] estimated the length of the cohesive zone as a function of the crack growth velocity. The expressions that result from these models for the case of plane stress are presented in Table 2. The relation between the critical stress intensity factor K_c and the critical energy release rate G_c can be expressed as $K_c^2 = G_c E$.

All of the models described above to predict the cohesive zone length l_{cz} have the form:

$$l_{cz} = ME \frac{G_c}{(\tau^0)^2} \quad (8)$$

where E is the Young modulus of the material, G_c is the critical energy release rate, τ^0 is the maximum interfacial strength, and M is a parameter that depends on each cohesive zone model. The most commonly used models in the literature are Hillerborg's model [15] and Rice's model [32]. In these models, the parameter M is either close or exactly equal to unity. A summary of the different models commonly used in the literature, and the equivalent parameter M for plane stress are shown in Table 2. Rice's model is used in the analysis described in this paper.

For the case of orthotropic materials with transverse isotropy and assuming plane stress, the value of the Young's modulus in Eq. (8) is the transverse modulus of the material E_2 .

For delamination in slender bodies, the cohesive zone length is a material and structural property [33]. For constitutive models that prescribe non-zero tractions when the displacement jump is zero, the length of the cohesive zone can be estimated as [33]

$$l_{cz} = \left[E \frac{G_c}{(\tau^0)^2} \right]^{1/4} t^{3/4} \quad (9)$$

The constitutive model used here prescribes zero tractions when the displacement jump is zero. Under this condition, the length of the cohesive zone predicted by Eq. (9) will be smaller than the effective cohesive length [33].

In order to obtain accurate FEM results using CZM, the tractions in the cohesive zone must be represented properly by the finite element spatial discretization. The number of elements in the cohesive zone is

$$N_e = \frac{l_{cz}}{l_e} \quad (10)$$

where l_e is the mesh size in the direction of crack propagation.

Table 2
Length of the cohesive zone and equivalent value of the parameter M

	l_{cz}	M
Hui et al. [31]	$\frac{2}{3\pi} E \frac{G_c}{(\tau^0)^2}$	0.21
Irwin [30]	$\frac{1}{\pi} E \frac{G_c}{(\tau^0)^2}$	0.31
Dugdale [13], Barenblatt [14]	$\frac{\pi}{8} E \frac{G_c}{(\tau^0)^2}$	0.40
Rice [32], Falk et al. [21]	$\frac{9\pi}{32} E \frac{G_c}{(\tau^0)^2}$	0.88
Hillerborg et al. [15]	$E \frac{G_c}{(\tau^0)^2}$	1.00

When the cohesive zone is discretized by too few elements, the distribution of tractions ahead of the crack tip is not represented accurately. Therefore, a minimum number of elements, N_c , is needed in the cohesive zone to get successful FEM results.

However, the minimum number of elements needed in the cohesive zone is not well established: Moës and Belytschko [34], based on the work of Carpinteri et al. [35], suggested using more than 10 elements. However, Falk et al. [21] used between 2 and 5 elements in their simulations. In the parametric study by Dávila and Camanho et al. [36], the minimum element length for predicting the delamination in a double cantilever beam (DCB) specimen was 1 mm, which leads, using Eq. (8) with $M = 1$, to a length of the cohesive zone of 3.28 mm. Therefore, three elements in the cohesive zone were sufficient to predict the propagation of delamination in mode I.

2.3. Guidelines for the selection of the parameters of the interface with coarser meshes

One of the drawbacks in the use of cohesive zone models is that very fine meshes are needed to assure a reasonable number of elements in the cohesive zone. The length of the cohesive zone given by Eq. (8) is proportional to the fracture energy release rate (G_c) and to the inverse of the square of the interfacial strength τ^0 . For typical graphite–epoxy or glass–epoxy composite materials, the length of the cohesive zone is smaller than one or two millimeters. Therefore, according to Eq. (10), the mesh size required in order to have more than two elements in the cohesive zone should be smaller than half a millimeter. The computational requirements needed to analyze a large structure with these mesh sizes may render most practical problems intractable.

Alfano and Crisfield [8] observed that variations of the maximum interfacial strength do not have a strong influence in the predicted results, but that lowering the interfacial strength can improve the convergence rate of the solution. The result of using a lower interfacial strength is that the length of the cohesive zone and the number of elements in the cohesive zone increase. Therefore, the representation of the softening response of the fracture process ahead of the crack tip is more accurate with a lower interface strength although the stress distribution in the regions near the crack tip might be altered [8].

It is possible to develop a strategy to adapt the length of the cohesive zone to a given mesh size. The procedure consists of determining the value $\bar{\tau}^0$ of the interfacial strength required for a desired number of elements (N_c^0) in the cohesive zone. From Eqs. (8) and (10), the required interface strength is

$$\bar{\tau}^0 = \sqrt{\frac{9\pi E G_c}{32 N_c^0 l_c}} \quad (11)$$

Finally, the interfacial strength is chosen as

$$T = \min\{\tau^0, \bar{\tau}^0\} \quad (12)$$

The interfacial strength is computed for each loading mode by substituting the values of the fracture toughness in Eq. (11) and of the ultimate traction in Eq. (12) by the values corresponding to the loading mode.

The effect of a reduction of the interfacial strength is to enlarge the cohesive zone, and thus, the model is better suited to capture the softening behaviour ahead of the crack tip. Although the stress concentrations in the bulk material near the crack tip are less accurate when using a reduced interfacial strength value, the mechanics of energy dissipation are properly captured, which ensures the proper propagation of the crack front. Moreover, if Eq. (7) is used to compute the interface stiffness, the interface stiffness will be large enough to assure a stiff connection between the two neighboring layers and small enough to avoid spurious oscillations.

The strategy of lowering the interfacial strength whilst keeping the fracture toughness constant was also proposed by Bažant and Planas [37] in a different context. The procedure developed by Bažant and Planas is used in the numerical implementation of crack band models where the energy dissipated is a function of the volume of the finite element [37]. In this case, it is necessary to modify the constitutive model including a characteristic element length to assure that the computed fracture energy is independent of the discretization. The reduction of the maximum stress is used to avoid snap-back of the modified constitutive model of large elements.

Using cohesive elements there is no need to use a characteristic element length in the constitutive model. It is however necessary to have a sufficient number of elements in the cohesive zone to capture the correct distribution of tractions. Therefore, the critical dimension of the finite element is along the crack propagation path.

When using crack band models in the simulation of delamination using coarse meshes, the strength must be adjusted taking into account the dimension of the element in the direction perpendicular to the crack plane (Bažant's model [37]), and also the dimension of the element in the crack propagation path using the method proposed here.

3. FEM implementation

The cohesive zone model used here is based on the model that was previously proposed by the authors [11,12]. The model has been adapted to be used in 2D finite element models. A brief description of the model, with special focus on the kinematics and constitutive damage model, is presented.

3.1. Kinematics and constitutive relation of cohesive zone models

The displacement jump across the interface of the material discontinuity required in the constitutive model, $\llbracket u_i \rrbracket$, can be obtained as a function of the displacement of the points located on the top and bottom side of the interface, u_i^+ and u_i^- , respectively:

$$\llbracket u_i \rrbracket = u_i^+ - u_i^- \quad (13)$$

where $u_i^\pm$ are the displacements with respect to the fixed Cartesian coordinate system. A co-rotational formulation is used in order to express the components of the displacement jumps with respect to the deformed interface. The coordinates $\bar{x}_i$ of the deformed interface can be written as [38]

$$\bar{x}_i = X_i + \frac{1}{2}(u_i^+ + u_i^-) \quad (14)$$

where X_i are the coordinates of the undeformed interface.

The components of the displacement jump tensor in the local coordinate system on the deformed interface, Δ_m , are expressed in terms of the displacement field in global coordinates:

$$\Delta_m = \Theta_{mi} \llbracket u_i \rrbracket \quad (15)$$

where Θ_{mi} is the rotation tensor.

The constitutive operator of the interface, D_{ji} , relates the element tractions, τ_j , to the displacement jumps, Δ_i :

$$\tau_j = D_{ji} \Delta_i \quad (16)$$

Under single-mode loading, controlled energy dissipation is achieved assuring that the area under the traction–displacement jump relation equals the corresponding fracture toughness, as illustrated in Fig. 3.

Under mixed-mode loading, a criterion using an interaction between the components of the energy release rate needs to be used to predict crack propagation.

The constitutive damage model used here, formulated in the context of Damage Mechanics (DM), was previously proposed by the authors [11,12]. All the details of the constitutive model are presented in Refs. [11,12] and will not be repeated here. The formulation presented in previous references is adapted to use the cohesive elements together with 2D plane strain elements.

The constitutive model prevents interpenetration of the faces of the crack during load-reversal, and uses a Fracture Mechanics-based criterion to predict crack propagation. The free energy density ψ is

$$\psi(\Delta, d) = (1 - d)\psi^0(\Delta_i) - d\psi^0(\delta_{ij}\langle -\Delta_i \rangle) \quad (17)$$

where ψ^0 is the free energy per unit surface, δ_{ij} is the Kronecker delta, d is a scalar damage variable, and $\langle \cdot \rangle$ is the MacAuley bracket defined as $\langle x \rangle = \frac{1}{2}(x + |x|)$.

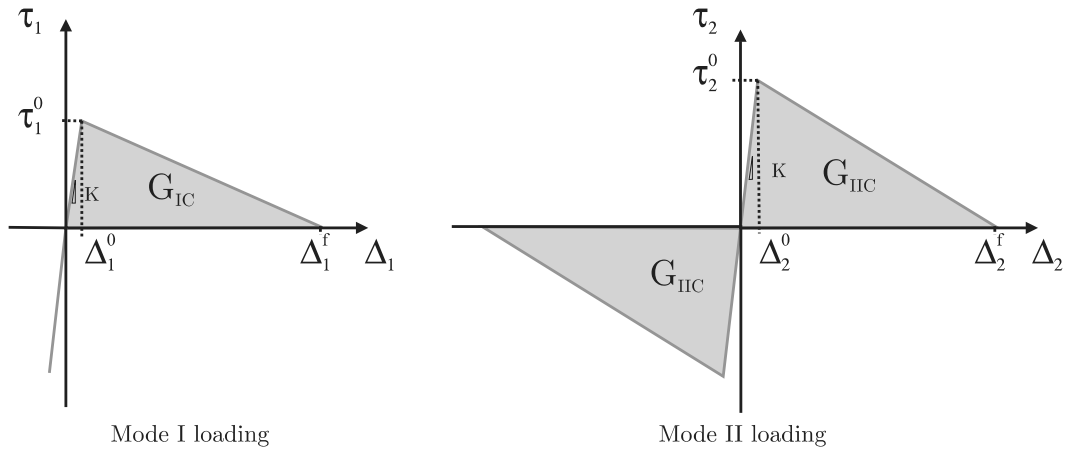


Fig. 3. Constitutive equations under mode I and mode II loading.

The constitutive equation obtained by differentiating the free energy with respect to the displacement jumps is

$$\tau_i = \frac{\partial \psi}{\partial \Delta_i} = (1 - d)\delta_{ij}K\Delta_j - d\delta_{ij}K\delta_{1j}\langle -\Delta_1 \rangle \quad (18)$$

A norm of the displacement jump tensor, denoted as λ , is used to compare different stages of the displacement jump states so that it is possible to define such concepts as ‘loading’, ‘unloading’ and ‘reloading’. The equivalent displacement jump is a non-negative and continuous function, defined as

$$\lambda = \sqrt{\langle \Delta_1 \rangle^2 + \Delta_2^2} \quad (19)$$

Δ_1 is the displacement jump in mode I, i.e., normal to midplane, and Δ_2 is the displacement jump in mode II. The damage criterion, formulated in the displacement jump space, is

$$\bar{F}(\lambda^t, r^t) := G(\lambda^t) - G(r^t) \leq 0 \quad \forall t \geq 0 \quad (20)$$

where $G(\cdot)$ is a suitable monotonic scalar function, ranging from 0 to 1 that governs the evolution of damage, and r^t is the damage threshold for the current time, t :

$$r^t = \max\{r^0, \max_s \lambda^s\}, \quad 0 \leq s \leq t \quad (21)$$

where r^0 denotes the initial damage threshold.

The evolution law for the damage threshold and the damage variable is defined as

$$\dot{d} = \dot{\mu} \frac{\partial \bar{F}(\lambda, r)}{\partial \lambda} = \dot{\mu} \frac{\partial G(\lambda)}{\partial \lambda}; \quad \dot{r} = \dot{\mu} \quad (22)$$

where $\dot{\mu}$ is a damage consistency parameter used to define loading–unloading conditions according to the Kuhn–Tucker relations:

$$\dot{\mu} \geq 0; \quad \bar{F}(\lambda^t, r^t) \leq 0; \quad \dot{\mu} \bar{F}(\lambda^t, r^t) = 0 \quad (23)$$

Further details regarding the damage model used here can be found in Refs. [11,12].

4. Simulation of the double cantilever beam specimen

The influence of mesh size, interface stiffness, interface strength, and the selection of the parameters of the constitutive equation according to the proposed methodology were investigated by analyzing the mode I delamination test (double cantilever beam, DCB). The DCB specimen was fabricated with a unidirectional

Table 3
Mechanical and interface material properties of T300/977-2 [10,39]

E_{11}	$E_{22} = E_{33}$	$G_{12} = G_{13}$	G_{23}
150.0 GPa	11.0 GPa	6.0 GPa	3.7 GPa
$\nu_{12} = \nu_{13}$	ν_{23}	G_{IC}	τ_3^0
0.25	0.45	0.352 N/mm	60 MPa

T300/977-2 carbon-fiber-reinforced epoxy laminate. The specimen was 150-mm-long, 20.0 mm-wide, with two 1.98-mm-thick arms, and it had an initial crack length of 55 mm. The material properties are shown in Table 3 [10].

The FEM model is composed of two layers of four-node 2D plane strain elements connected together with four-node cohesive elements. The cohesive elements were implemented using a user-written subroutine in the finite element code ABAQUS [40].

Several sets of simulations were performed. A very refined mesh using an element length of 0.0125 mm was used in the simulation of a DCB specimen to predict the distribution of tractions ahead of the crack tip and the corresponding length of the cohesive zone.

The distribution of tractions ahead of the crack tip at the peak load of DCB specimen is shown in Fig. 4.

The length of the cohesive zone obtained from the analysis results shown in Fig. 4 is 0.9 mm. Using the material parameters shown in Table 3 and Eq. (8), the parameter M is calculated as 0.84. This value is closest to Rice's model (0.88), shown in Table 2.

Other DCB tests were simulated with different levels of mesh refinement using the material properties shown in Table 3 and interfacial stiffness of $K = 10^6$ N/mm³. Eqs. (7) and (10) were used to calculate an adjusted interfacial strength and interface stiffness. Finally, a set of simulations with a constant mesh size using different interface stiffnesses were performed in order to investigate the influence of the stiffness on the calculated results.

To study the effects of mesh refinement, several analyses were carried out for element sizes ranging between 0.125 mm and 5 mm. The corresponding load–displacement curves are shown in Fig. 5. The results indicate that a mesh size of $l_e \leq 0.5$ mm is necessary to obtain converged solutions. The predictions made with coarser meshes overpredict significantly the experimental results. The length l_{cz} of the cohesive zone for the material

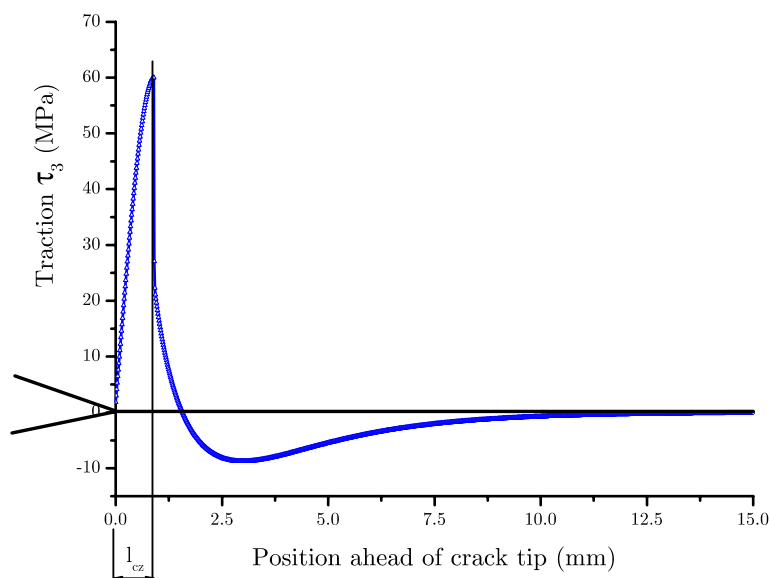


Fig. 4. Distribution of tractions ahead of the crack tip at peak load.

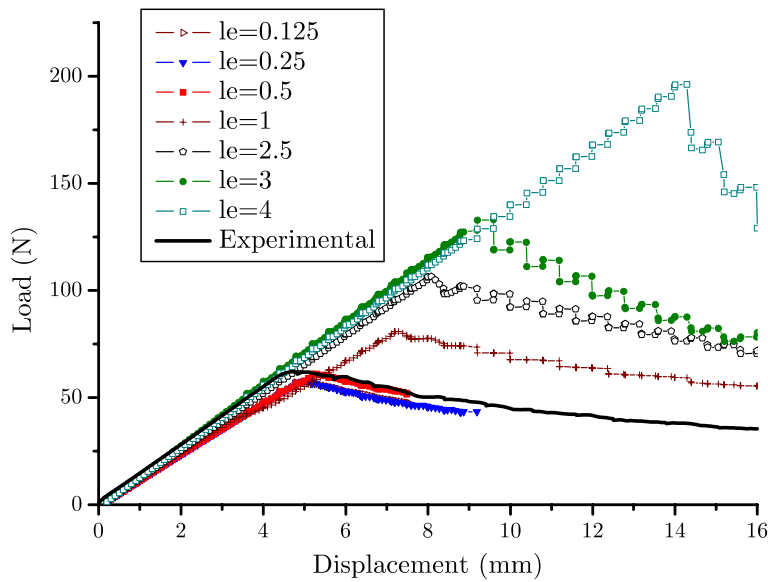


Fig. 5. Load–displacement curves using the nominal interface strength ($\tau^0 = 60$ MPa) for a DCB test with different mesh sizes.

given in Table 3 is 0.95 mm. Therefore, for a mesh size greater than 0.47 mm, fewer than two elements would span the cohesive zone, which is not sufficient for an accurate representation of the fracture process [35,36].

The error obtained from using a coarse mesh in the simulation of delamination in a DCB test specimen is related with the representation of the contact tractions ahead of the crack tip. The predicted distributions of the normal tractions ahead of the crack tip at 30 kN using a fine mesh (element length of 0.0125 mm) and a coarse mesh (element length of 1 mm) are shown in Fig. 6.

For an applied load of 30 kN, the fine-mesh model captures the softening region ahead of the crack tip while the coarse-mesh model does not.

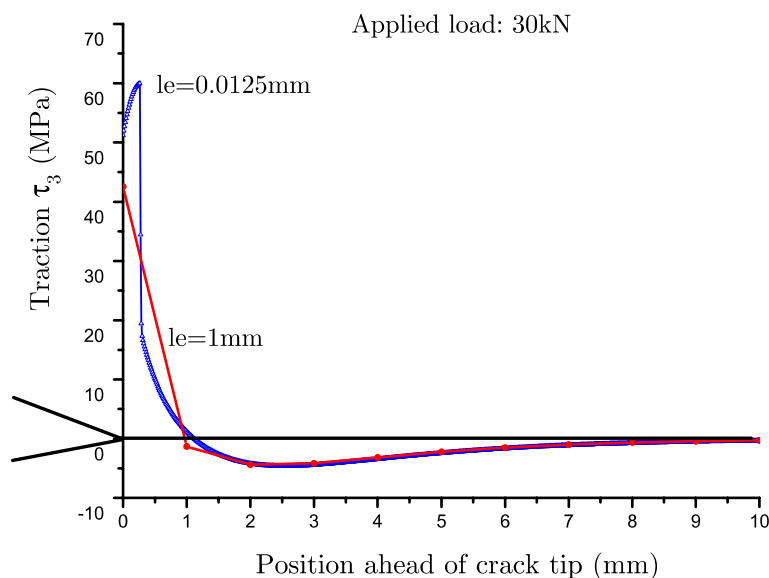


Fig. 6. Distribution of tractions ahead of the crack tip for an applied load of 30 kN.

4.1. Effect of interface strength

To verify the effect of the interface strength on the numerical results, simulations were performed by specifying the desired number of elements within the cohesive zone to be $N_0 = 5$ and reducing the interface strength according to Eq. (12). The load–displacement curves obtained for several levels of mesh refinement are shown in Fig. 7. Accurate results are obtained for mesh sizes smaller than 2.5 mm.

A comparison of the maximum loads predicted using the nominal interface strength and the strength obtained from Eq. (12) is shown in Fig. 8. The failure load obtained by keeping the maximum interfacial

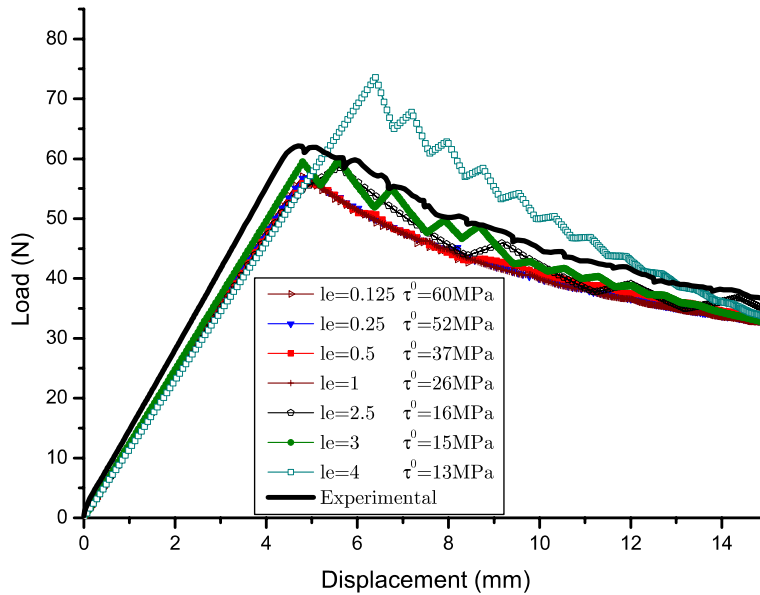


Fig. 7. Load–displacement curves obtained for a DCB test with different mesh sizes with the interface strength modified to keep $N_e \geq 5$.

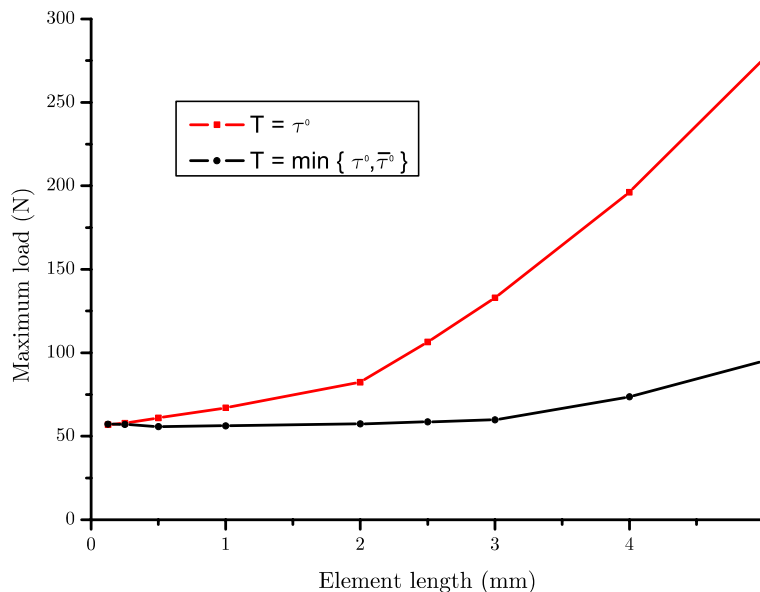


Fig. 8. Maximum load obtained in a DCB test for two cases: (a) with constant interfacial strength, (b) with interfacial strength calculated according to Eq. (12).

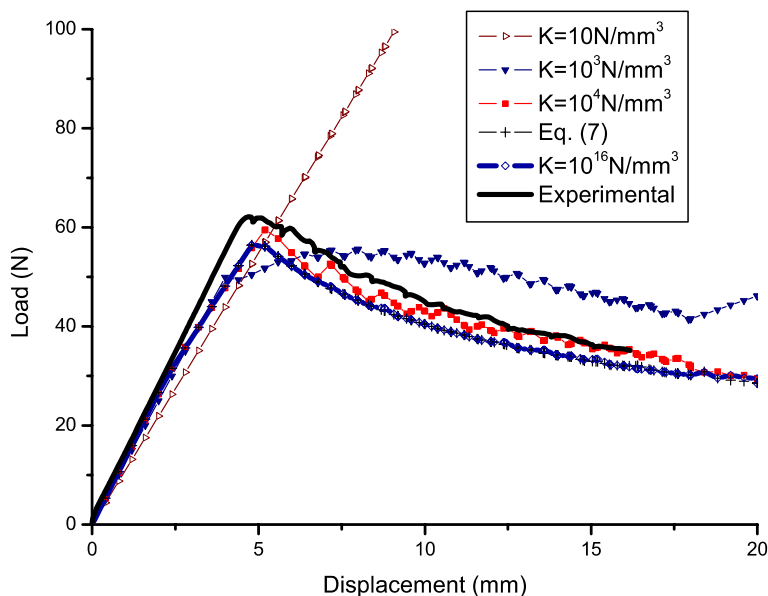


Fig. 9. Influence of the interface stiffness on the load–displacement curves.

strength constant increases with the mesh size. Mesh dependence is especially strong for mesh sizes greater than 2 mm. However, the failure loads predicted by modifying the interfacial strength according to Eq. (12) are nearly constant for element sizes smaller than 3 mm. In addition, the finite element results shown that the global deformation and the crack tip position are also nearly independent from mesh refinement.

4.2. Effect of interface stiffness

The DCB test was simulated with a mesh size of 2.5 mm for various values of the interface stiffness in order to investigate the influence of the stiffness on the predicted failure load. The results of the simulations are presented in Fig. 9.

The load–displacement response curves obtained from simulations using an interface stiffness greater than 10^4 N/mm^3 are virtually identical. However, smaller values of the interface stiffness have a strong influence on the load–displacement curves, since a stiff connection between the two neighboring layers is not assured. Moreover, the number of iterations needed for the solution when using an interface stiffness smaller than 10^4 N/mm^3 is greater than the number of iterations needed for a range of the interface stiffness between 10^6 and 10^{10} N/mm^3 . For values of the interface stiffness significantly greater than 10^{10} N/mm^3 , the number of iterations needed for the solution increases. The stiffness that results from Eq. (7) with $\alpha = 50$ is $5.55 \times 10^5 \text{ N/mm}^5$, which is ideal for a good convergence of the solution procedure.

5. Simulation of free-edge delamination

Free-edge delamination is a typical mode of damage initiation in specimens without pre-existing cracks [41–43]. To examine the effect of mesh size on a problem in which crack initiation is the critical issue, analyses were performed to predict the initiation of free edge delaminations of $[\pm 25^\circ/90^\circ]_s$ laminates studied by Crossman et al. [42].

5.1. Problem statement

The $[\pm 25^\circ/90^\circ]_s$ graphite–epoxy laminates investigated by Crossman were subjected to uniaxial strain. The laminates were fabricated from Fiberite T300/934 prepreg tape using an autoclave. The cross-section of the

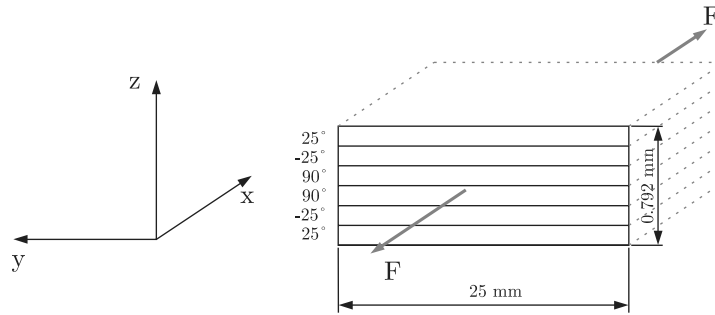


Fig. 10. Cross-section of the laminate.

Table 4
Mechanical properties of T300/934 graphite–epoxy [41,44]

E_{11}	$E_{22} = E_{33}$	$G_{12} = G_{13}$	G_{23}	$\nu_{12} = \nu_{13}$	ν_{23}
140.0 GPa	11.0 GPa	5.5 GPa	3.61 GPa	0.29	0.52

specimen was 25.0-mm-wide and 0.792-mm-thick (see Fig. 10). No initial cracks were induced in the laminate. The material properties are shown in Table 4 [41,44].

Tensile tests were performed by applying a controlled displacement in the x -direction. Due to the stacking sequence, the through the thickness free-edge stress distribution produces high σ_{zz} in the 90° ply. Therefore, high tractions occur at the $90^\circ/90^\circ$ interface, leading to the onset of delamination at the free-edge.

5.2. Numerical predictions

A FEM model was developed using six layers of four-node generalized plane strain elements [45,46]. The generalized plane strain model developed calculates the out-of-plane components of the stress tensor that cause delamination onset. The $90^\circ/90^\circ$ interface was modeled with the four-node cohesive elements presented in previous sections. The elastic properties of each layer were defined by means of the 3D stiffness matrix. The

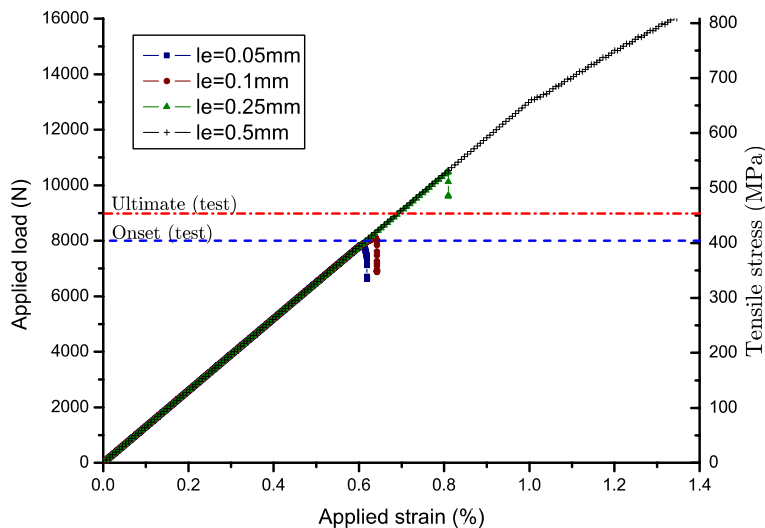


Fig. 11. Load–displacement curves obtained for a free-edge test with different mesh sizes using nominal interface strength properties.

cohesive elements were implemented using a user-written subroutine in the finite element code ABAQUS [40]. Due to the symmetry of the laminate, only half of the cross-section was modeled.

Two sets of simulations were performed. First, tensile tests were simulated with different levels of mesh refinement. Then, Eqs. (10) and (7) were used to calculate the interface strength and the interface stiffness.

Several analyses were carried out for mesh sizes ranging between 0.05 mm and 0.5 mm. The load–displacement curves obtained are shown in Fig. 11. An interfacial stiffness of $K = 10^6$ N/mm³, an interface strength of 51.7 MPa, a critical energy release rate of 0.175 N/mm [41], and the material properties shown in Table 4 were used. The length of the cohesive zone obtained with these properties, using Eq. (8), is 0.63 mm. A mesh size

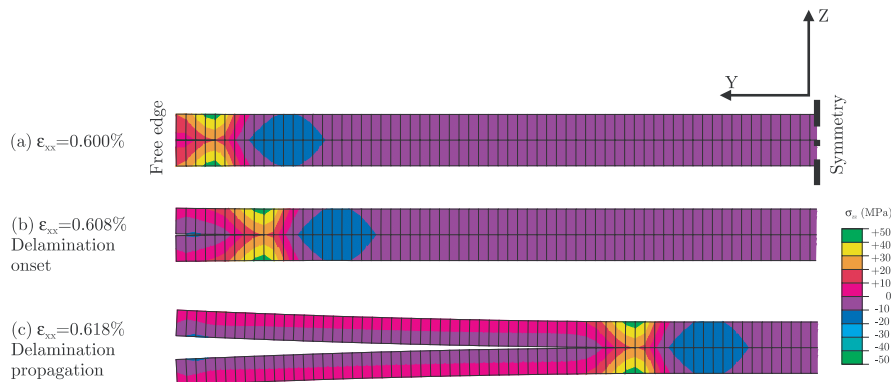


Fig. 12. Evolution of stresses during delamination onset and propagation of the two interior layers of the laminate.

Table 5

Summary of predicted and experimental values of tension tests on T300/934 laminates

	Predicted	Experimental	Difference (%)
Strain at delamination onset (%)	0.6	0.59	2
Stress at delamination onset (MPa)	392	404	−3
Ultimate strain (%)	0.6183	0.66	−6
Ultimate strength (MPa)	409	459	−10

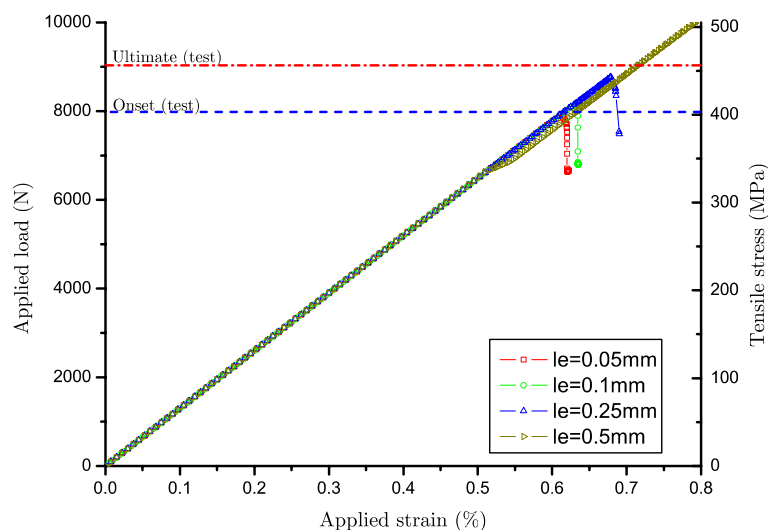


Fig. 13. Load–displacement curves obtained for a free-edge test with different mesh sizes and with the interface strength adjusted for $N_c = 5$.

smaller than 0.2 mm is needed in order to have three or more elements in the cohesive zone. The predictions obtained using different levels of mesh refinement are shown in Fig. 11. Using meshes coarser than 0.2 mm, the correct solution is significantly overpredicted. For a mesh size equal to 0.5 mm the maximum applied load is greater than two times the maximum load obtained with meshes smaller than 0.1 mm.

The ultimate tensile strength calculated using a mesh size of 0.1 mm is 404 MPa. Delamination onset occurs when a point in the interface is not able to carry any traction, which changes the stress distribution. The stress distribution of the two interior layers of the laminate in a plane perpendicular to the interface and normal to the load direction are represented in Fig. 12 for three stages of deformation. At the first stage, before delamination onset, the stresses σ_{zz} near the free edge are tensile due to the mismatch effect of the Poisson ratio (red-lighter zones in Fig. 12). At delamination onset, a region of the free edge is unable to carry load and the

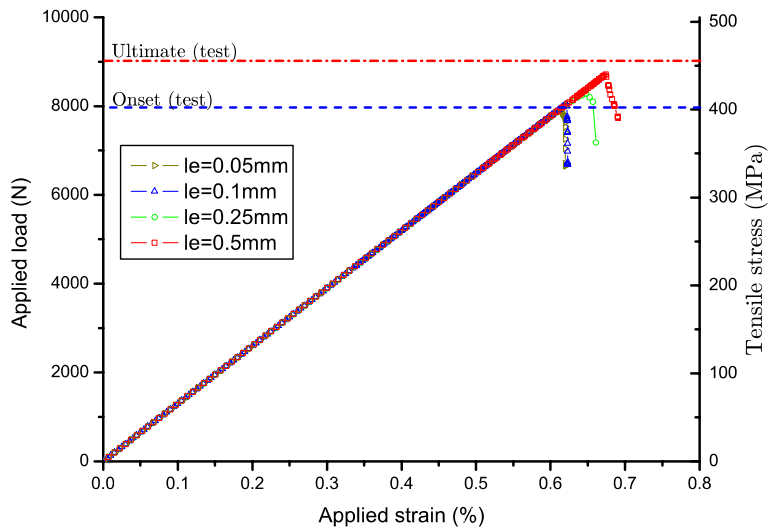


Fig. 14. Load-displacement curves obtained for a free-edge test with different mesh sizes and with the interface strength adjusted for $N_e = 10$.

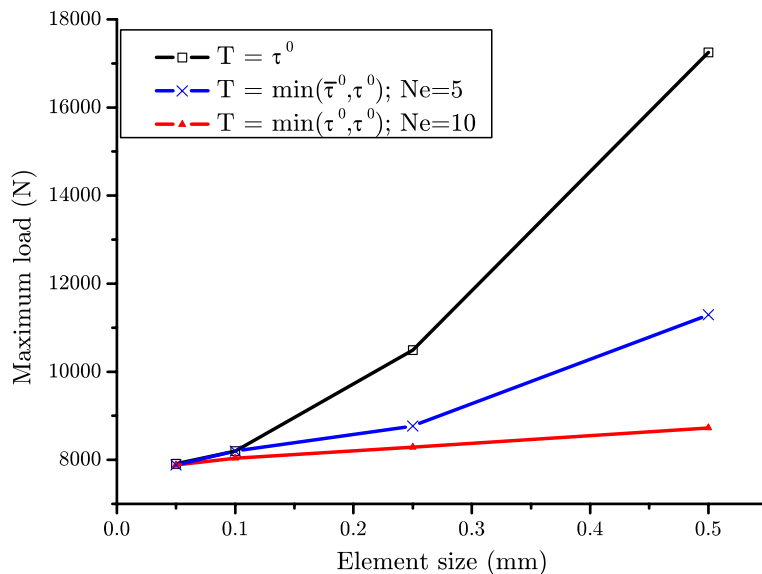


Fig. 15. Maximum load obtained in a free-edge test for three cases: (a) with constant interfacial strength, (b) and (c) with interfacial strength calculated according to Eq. (12) with N_e equal to 5 and 10.

stresses become compressive, as shown in Fig. 12b (blue-darker zones). The predicted onset of delamination is $\varepsilon_{xx} = 0.6\%$, and the corresponding stress is 392 MPa. The experimental results obtained by Crossman and Wang [42] reported a delamination onset tensile stress of 409 MPa and an ultimate tensile strength of 459 MPa. The predicted and experimental onset and failure stresses and strains are summarized in Table 5. Unstable delamination propagation and the corresponding structural collapse occurs at a strain of 0.6183%.

To verify the effect of interface strength on the predicted results, simulations were performed by specifying the desired number of elements spanning the cohesive zone to be $N_e = 5$ and reducing the interface strength according to Eq. (12). The load–displacement curves obtained for several levels of mesh refinement are shown in Fig. 13.

A similar study of mesh size effect was repeated by selecting the desired number of elements spanning the cohesive zone to be $N_e = 10$. Although the results presented in Figs. 14 and 15 for $N_e = 10$ are more accurate than those for $N_e = 5$, the improvement is insignificant when the element is smaller than 0.25 mm.

Figs. 14 and 15 show that the maximum load predicted by the numerical model converges to the same value when decreasing the element length. The maximum load predicted differs 12% of the experimental value. This difference can be justified by the increase of the fracture toughness associated with fiber bridging that is not taken into account in the model.

6. Concluding remarks

An engineering solution for the simulation of delamination using coarse meshes was developed. Two new guidelines for the selection of the parameters for the constitutive equation used for the simulation of delamination were presented.

First, a new equation for the selection of the interface stiffness parameter K was derived. The new equation is preferable to previous guidelines because it results from mechanical considerations rather than from experience. The approach provides an adequate stiffness to ensure a sufficiently stiff connection between two neighboring layers, while avoiding the possibility of spurious oscillations in the solution caused by overly stiff connections.

Finally, an expression to adjust the maximum interfacial strength used in the computations with coarse meshes was presented. It was shown that a minimum number of elements within the cohesive zone is necessary for accurate simulations. By reducing the maximum interfacial strength, the cohesive zone length is enlarged and the cohesive zone spans more elements. The results obtained by reducing the maximum interfacial strength show that accurate results can be obtained with a mesh ten times coarser than by using the nominal interface strength. The drawback in using a reduced interfacial strength value is that the stress concentrations in the bulk material near the crack tip are less accurate.

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